SCHULICH School of Engineering

COURSE OUTLINE
FALL 2026

Approval date	Sept 10 2026
Approved by	DTW
Approver Department	ESE

ENEL 500A – Computer, Electrical, and Software Engineering Team Design

1. Course Description

Preliminary and detailed engineering design and implementation of an engineering system that applies engineering knowledge to solving a real-life problem. The emphasis is on the design process as it is associated with electrical, computer and software engineering, design methodology, general design principles for engineers, teamwork and project management.

Prerequisite(s): Fourth year standing or above.

Antirequisite(s): Credit for Electrical Engineering 500 and either 583 or 589 will not be allowed.

Notes: Concurrent enrolment in Electrical Engineering 500 and one or more of Internships 513.01, 513.02, 513.03, and 513.04 will not be allowed.

Course Hours: 6 units; F(1-3)

Academic Credits: 6

Calendar Reference: https://calendar.ucalgary.ca/courses/1630001

2. Timetable

Section	Days of the Week	Time	Location
LEC 01	R	18:30-19:20	ST 140
LAB B01	M	17:00-19:50	ENG 24
LAB B02	T	17:00-19:45	ENG 24

3. Course Instructors

Name: Hamid Zareipour
Section: 01
Email: hzareipo@ucalgary.ca
Office: ENA 232
Phone: 210-9516

Teaching Assistants

Section	First Name	Family Name	Email
LEC01	Basit Oladipupo	Olatunbosun	basit.olatunbosun@ucalgary.ca
LEC01	Shashank Suryakant	Pantaone	shashank.pantaone@ucalgary.ca
LEC01	Yumnam Sudipta	Devi	yumnam.devi@ucalgary.ca
LEC01	Samuel	Owyusu	samuel.owusu1@ucalgary.ca
LEC01	Desiree	Leal	desiree.leal1@ucalgary.ca
LEC01	Youhao	Huang	youhao.huang@ucalgary.ca
LEC01	Ali	Forootani	ali.forootani1@ucalgary.ca
LEC01	Mohammad	alhashem	mohammad.alhashem@ucalgary.ca
LEC01	Md Afif Al	Mamun	mdafifal.mamun@ucalgary.ca
LEC01	Samuel	Wang	zirui.wang1@ucalgary.ca
LEC01	Mohsen	Tavakolian	mohsen.tavakolian@ucalgary.ca
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LEC01	Reyhaneh	Farahmand	reyhaneh.farahmand@ucalgary.ca
LEC01	Mutahhar	Chishti	mutahhar.chishti@ucalgary.ca

4. Learning Outcomes

Upon successful completion of the course, you will be able to:

1. Analyse an engineering problem. **(GA2)**

2. Investigate and develop alternative solutions for an engineering problem. **(GA3)**

3. Apply the principles of engineering design to solve an engineering problem. **(GA4)**

4. Work as part of a team and contribute to an engineering project as an individual as well as a team member. **(GA6)**

5. Apply communication skills towards the end goal of successfully completing an engineering project. **(GA7)**

6. Understand the importance of professionalism in a team project. **(GA8)**

7. Apply the principles of economics and project management in a team design project. **(GA11)**

8. Explore and learn the academic background needed for a new engineering project. **(GA12)**

Graduate attributes

The bracketed references (e.g., **GA 1**) identify the Canadian Engineering Accreditation Board (CEAB) Graduate Attributes that are developed and assessed through each course learning outcome. The Graduate Attributes listed below describe the knowledge, skills, and professional competencies expected of graduates from accredited Canadian engineering programs. Together, they help ensure that engineering graduates are prepared for professional practice.

GA1. A knowledge base for engineering
GA2. Problem Analysis
GA3. Investigation
GA4. Design
GA5. Use of engineering tools
GA6. Individual and team work
GA7. Communication skills
GA8. Professionalism
GA9. Impact of engineering on society/environment
GA10. Ethics and equity
GA11. Economics and project management
GA12. Life-long learning

5. Course Evaluation

Assessment Components

Component	**Due Date**	**Submission Type**	**Learning Outcomes**	**Weight**
Initial Project Proposal	Sept 13	Team	1	Pass/Fail
TA Kick-off Meeting	Sept 21- 29	Team	4, 5, 6	Pass/Fail
Finalized Project Proposal	Oct 4	Team	1	2%
Team Contract and Sponsor-Team Contract	Oct 4	Team	4	1.5%+1.5%

Project Information (Google Form) **includes finalizing your academic advisor/Industry Rep*	Oct 12	Team	5	Pass/Fail
Team Affiliation (Google Form)	Oct 12	Individual	5	Pass/Fail
Cover Page PDFs to Dropbox	Oct 12	Team	5	Pass/Fail
Design Roadmap Document **includes finalizing your technician for hardware projects*	Nov 8	Team	1 - 7	16%
Fall Term Progress Evaluations (Three Evaluators: TA, Academic Advisor, Sponsor Rep)	Nov 18 - Dec 2	Team	5	7%+7%+7%
Peer Evaluation	Dec 6	Individual	4	Pass/Fail
Winter Term Progress Evaluation (Three Evaluators: TA, Academic Advisor, Sponsor Rep)	Jan 25 - Feb 5	Team	1 - 7	3%+3%+3%
Project Demonstration Final Evaluations (Three Evaluators: TA, Academic Advisor, Sponsor Rep)	Mar 15 - Apr 2	Team	1 - 7	10%+10%+10%
Design Fair Poster and Showcasing	TBA	Team	5	Pass/Fail
Final Report	Apr 12	Team	1 – 7	7%
Peer Evaluation 2	Apr 19	Individual	4	12%

There are no exams in this course. All evaluations except for PEER Evaluation 2 are team evaluations. There is flexibility in the timing of all submissions. They all happen over a period of time. All students must attend all team evaluations. Thus Teams are required to schedule the evaluations accordingly. Exceptional cases, including missed evaluations, must be discussed with the course coordinator and will be addressed case by case.

a) It is necessary to earn a passing grade on all grade components in order to pass the course as a whole. The passing grade is 60% of the weight of the component.

b) Submission deadline hour is 11:59 PM of the specified date, unless specified otherwise. Where applicable, submissions within 2 to 12 hours after the deadline will automatically get a 15% penalty; submissions within 12 and 24 hours after the deadline will get a 25% penalty; and, submissions will be graded as zero after 24 hours passed the deadline.

c) There are several items in the marking scheme table that will be given a pass or a fail. For items that are the responsibility of individual students, each fail will result in the deduction of 4% out of the total 100% final numerical grade. For team items with a pass/fail mark, failing to do each of the specified tasks associated with each item will result in the deduction of 4% out of the total 100% final numerical grade from all team members.

d) In exceptional circumstances, the instructor may require individual students or teams to complete additional assignments, attend meetings or participate in activities. Those will be communicated in advance. They will be graded with a Pass/Fail.

e) Attendance Rules: All students must be present in evaluations sessions. For other course activities, the following general attendance rule will apply. Attendance will be taken per team. Only one student from the team can be absent. That student must email the instructor and the TA and cc all team members and justify why he/she will be missing a team activity. Each unapproved absence, one Fail will be counted and rule explained in item (c) above will apply.

f) Individual Performance: All grading components outlined in the table of Section 7, above, are automatically applied to all team members, except for the peer evaluation part. However, if the instruction team has reasons to believe individual students have not equally participated in the project, any of the grading components may be scaled down for those individuals. The scaled down individual grade may lead to a failure in the course if any of the final grade components do not meet the minimum passing requirement. The instruction team may have to withhold some information in such circumstances to ensure privacy and confidentiality principles.

g) Attending the final design fair is mandatory for all teams. At least one student must be present at any time during the fair. Failing to showcase the project in the fair will automatically result in failing the course.

Final Grade Determination

The final grade in this course will be based on the components listed in Assessment Components table, above.

Conversion from a score out of 100 to a letter grade will be done using the conversion chart shown below. This grading scale can only be changed during the term if the grades will not be lowered.

Letter Grade	Lower Bound	Upper Bound
A+	99%	
A	97%	99%
A-	93%	97%
B+	90%	93%
B	85%	90%
B-	80%	85%
C+	75%	80%
C	70%	75%
C-	66%	70%
D+	63%	66%
D	60%	63%
F		60%

Notes:

Course Format and Scheduling

This course is not a conventional “lecture--based” course. Learning by doing is the model here. However, the instructor may require occasional class sessions that would benefit all students and teams. Such sessions will be announced in advance. You are encouraged to attend. Such sessions will be recorded and posted on D2L. Thus, unless it is announced on the course D2L page, there will be no lecture sessions. The allocated times and venues for the lecture and lab are meant to provide all students with available time slots and locations in case they need to get together and work in person as a team.

This course requires students to work as a team. The plan for this academic year is to have labs and facilities available to students. However, things may change. The teams must have a plan for mitigating the risks imposed by the covid situation. Also, it is between the team members to have a procedure/plan for in-person or online meetings/teamworks and how to handle special cases (for example, if a member is not vaccinated, or is remote, or mask policies for the team). The university policies as announced publicly are the basis for making those plans. The instructor will not be mandating any solutions in special cases other than what is mandated by the University of Calgary. However, the teams are welcome to consult their situation with the TAs or the instructor.

Please regularly visit the course D2L web page for updates as the course unfolds. Tasks, their descriptions, rubrics, expectations and all other information \ instructions that you need on how to make your project a success will be posted there.

6. Learning Resources

Textbook(s)

Recommended Textbooks:
Title: Engineering Project Management
Author(s): N. J. Smith
Edition: Third
Publisher: Backwell Publishing

Title: Engineering Design
Author(s): Clive L, Dym and Patrick Little
Edition: Third
Publisher: Wiley

Technology Requirements

None

7. Class Schedule

This course is not a conventional "lecture--based" course. Learning by doing is the model here. However, the instructor may require occasional class sessions that would benefit all students and teams. Such sessions will be announced in advance. You are encouraged to attend. Such sessions will be recorded and posted on D2L. Thus, unless it is announced on the course D2L page, there will be no lecture sessions. The allocated times and venues for the lecture and lab are meant to provide all students with available time slots and locations in case they need to get together and work in person as a team.

8. Course Policies and Expectations

8.1. Use of AI tools

Transparent Use: AI tools may be used to support your learning and content creation, with appropriate attribution. You may use AI for drafting, problem-solving support, or code assistance, provided that you critically evaluate and revise all outputs. You are responsible for the accuracy and integrity of your submission. All use of AI, including prompts and tool identification, must be clearly documented through an ***AI use statement*** included with your work.

8.2. Use of Calculators in Examinations

You may use any calculator you wish for studying and completing lab reports. However, you must use only one of the following sanctioned Schulich School of Engineering calculators during exams: Casio FX-260S, Casio FX-300MS, Casio FX-95ES Plus, Casio FX-300ESPLUS2 (2nd Edition), TI-30XIIS, TI-30Xa, Sharp EL-531XTB-WH, and HP10S Plus.

8.3. Reappraisals

For reappraisals of graded term work or academic assessments (including final grades), please refer to the **SSE Reappraisal of Graded Term Work and Academic Assessments Policy**, which is available at:
https://schulich.ucalgary.ca/sites/default/files/teams/1/SSE%20Reappraisal%20Policy.pdf

Reappraisal requests must be based on an alleged grading error and follow the procedures and deadlines specified in the policy. For graded term work, students must first discuss their concerns with the course instructor within 10 business days of receiving or viewing the graded work. Reappraisal outcomes may result in a grade increase, decrease, or no change. Forms are available on the Engineering Student Center D2L site.

8.4. Attendance and Participation Expectations

Students are expected to attend class regularly and to be fully present and engaged in class activities and discussions. These are part of the participation grade, as outlined in the assessment components section above. If students miss a class session that is included as a component of participation, they must contact the instructor to discuss options to make-up for that component.

8.5. Missed Coursework

8.6. Conduct

Students are expected to demonstrate behaviour in class that promotes and maintains a positive and productive learning environment. This includes treating classmates, instructors, teaching assistants, staff, and external partners with courtesy and professionalism; engaging honestly in all academic activities; contributing constructively to discussions and team-based work; and respecting diverse perspectives.

As members of the University of Calgary community, students are expected to demonstrate conduct that is consistent with the University's Conduct and Non-Academic Misconduct policy and procedures: https://calendar.ucalgary.ca/uofcregs/university-regulations/integrity-conduct

As members of an engineering program, students are also expected to begin developing the professional behaviours and ethical judgment expected of engineers. Throughout this course, students should demonstrate integrity, accountability, respect for others, and a commitment to public safety and responsible engineering practice. Students are encouraged to familiarize themselves with the ethical responsibilities of the engineering profession as described in the Association of Professional Engineers and Geoscientists of Alberta (APEGA) Ethical Practice guideline:
https://www.apega.ca/docs/default-source/pdfs/standards-guidelines/ethical-practice.pdf

8.7. Use of Electronic Communication Devices in Class

The use of laptop and mobile devices in class is acceptable when used in a manner appropriate to the course and classroom activities. Please refrain from accessing websites and resources that may be distracting to you or for other learners during class time. Students are responsible for being aware of the University's Internet and email use policy, which can be found at

https://www.ucalgary.ca/legal-services/university-policies-procedures/acceptable-use-electronic-resources-and-information-policy

8.8. Expectations for Writing

All written assignments (including, to a lesser extent, written exam responses) will be assessed at least partly on writing skills. Writing skills include not only surface correctness (grammar, punctuation, sentence structure, etc.) but also general clarity and organization. Sources used in research papers must be properly documented. If you need help with your writing, you may use the **Writing Support** (https://ucalgary.ca/student-services/student-success/writing-support) at the **Chancellor Cuthbertson Student Success Centre** (https://www.ucalgary.ca/student-services/student-success/student-success). For further information, please refer to the official online University of Calgary Calendar, Academic Regulations, E. Course Information, E.2: Writing Across the Curriculum: https://calendar.ucalgary.ca/pages/2c2d1ce47b8c4d008aec9cc3da49876e

9. Guidelines for Recording Sessions

For more information about recording classes in any learning environment see the ***Media Recording of Students in Learning Spaces policy*** here: https://www.ucalgary.ca/provost/teaching-learning/recordings-learning-environments

9.1. Media Recording for Study Purposes

Students who wish to **audio record** lectures for personal study purposes need to follow the guidelines outlined in Section E.6 of the University Calendar: https://calendar.ucalgary.ca/pages/4ba66d5bd43b422ba2e9ab3d6331d5b0

Unless the audio recording of lectures is part of a student accessibility requirement, permission must be sought by the course instructor to audio record lectures.

9.2. Media Recording for Lesson Capture

The instructor may use media recordings to capture the delivery of a lecture. These recordings are intended to be used for lecture capture only and will not be used for any other purpose. Recordings will be posted on D2L for student use and will normally be deleted at the end of term. Students are responsible for turning off their camera and/or microphone if they do not wish to be recorded.

9.3. Media Recording for Assessment of Student Learning

The instructor may use media recordings as part of the assessment of students. This may include but is not limited to classroom discussions, presentations, clinical practice, or skills testing that occur during the course. These recordings will be used for student assessment purposes only and will not be shared or used for any other purpose. The recording will be destroyed as specified by Retention Rule 2000.01 “Examinations and Student Assignments.” https://www.ucalgary.ca/provost/teaching-learning/recordings-learning-environments

9.4. Media Recording for Self-Assessment of Teaching Practices

The instructor may use media recordings as a tool for self-assessment of their teaching practices. Although the recording device will be fixed on the instructor, it is possible that student participation in the course may be inadvertently captured. The recording will be destroyed as specified by Retention Rule 90.0011 “Draft Documents & Working Materials."
https://www.ucalgary.ca/provost/teaching-learning/recordings-learning-environments

10. University of Calgary Policies and Supports

10.1. SSE Advising and Policies

All Schulich School of Engineering students have access to a D2L site titled ***Engineering Student Centre***. Students have the responsibility to familiarize themselves with the policies available on this site.

10.2. Academic Misconduct

Academic Misconduct refers to student behavior which compromises proper assessment of a student’s academic activities and includes cheating; fabrication; falsification; plagiarism; unauthorized assistance; failure to comply with an instructor’s expectations regarding conduct required of students completing academic assessments in their courses; and failure to comply with exam regulations applied by the Registrar.

For information on the University of Calgary Student Academic Misconduct Policy and Procedure and the SSE Academic Misconduct Operating Standard, please visit:
https://schulich.ucalgary.ca/current-students/undergraduate/student-resources/policies-and-procedures

Additional information is available on the Academic Integrity Website at:
https://ucalgary.ca/student-services/student-success/learning/academic-integrity

10.3. Academic Accommodation

It is the student’s responsibility to request academic accommodations according to the University policies and procedures listed below. The Student Accommodations policy is available at:
https://ucalgary.ca/student-services/access/current-students/academic-accommodations

Students needing an accommodation because of a disability, or medical concerns should communicate this need to **Student Accessibility Services (SAS)** in accordance with the ***Procedure for Accommodations for Students with Disabilities*** available at: https://www.ucalgary.ca/legal-services/sites/default/files/teams/1/Policies-Accommodation-for-Students-with-Disabilities-Procedure.pdf. For additional information on support services and accommodations for students with disabilities, visit www.ucalgary.ca/access/

Students needing an accommodation in relation to their coursework or to fulfill requirements for a degree, based on a ***Protected Ground other than Disability***, should communicate this need by

submitting the ***SSE Request for Academic Accommodation Form*** to the Associate Head (Undergraduate Studies) of the department offering the course, within 10 business days prior to the class, test, exam, or assignment at issue.

10.4. Research Ethics

Students are advised that any research with human participants – including any interviewing (even with friends and family), opinion polling, or unobtrusive observation – must have the approval of the **Conjoint Faculties Research Ethics Board** (https://research.ucalgary.ca/conduct-research/ethics-compliance/human-research-ethics/conjoint-faculties-research-ethics-board-cfreb) or the **Conjoint Health Research Ethics Board** (https://research.ucalgary.ca/conduct-research/ethics-compliance/human-research-ethics/conjoint-health-research-ethics-board-chreb)

In completing course requirements, students must not undertake any human subjects research without discussing their plans with the instructor, to determine if ethics approval is required. Some courses will include assignments that involve conducting research with human participants; in these cases, the instructor will have applied for and received ethics approval for the course assignment. The instructor will discuss the ethical requirements for the assignment with the students.

For further information see Section E.5 Ethics of Human Studies of the University Calendar: https://calendar.ucalgary.ca/pages/627ed88eb4b041b7a2e8155effac3501

10.5. Instructor Intellectual Property

Course materials created by instructors (including presentations and posted notes, labs, case studies, assignments and exams) remain the intellectual property of the instructor. These materials may NOT be reproduced, redistributed or copied without the explicit consent of the instructor. The posting of course materials to third party websites such as note-sharing sites without permission is prohibited. Sharing of extracts of course materials with other students enrolled in the course at the same time may be allowed under fair dealing.

10.6. UCalgary Notice of Collection

The University of Calgary (UCalgary) respects your privacy and is committed to ensuring the privacy of all students, staff, and community members. Your personal information is being collected under the authority of section 4(c) of the ***Protection of Privacy Act*** (**POPA**). It will be collected, used and disclosed for the purposes outlined in UCalgary's Notice of Collection, Use and Disclosure of Personal Information (https://www.ucalgary.ca/legal-services/access-information-privacy/ucalgary-notice-collection) and managed in accordance with UCalgary's Privacy Management Program (https://www.ucalgary.ca/legal-services/access-information-privacy/privacy-management-program).

Your personal information may be input into automated systems to generate content or make decisions, recommendations, or predictions.

Assignments can only be returned to the student and will be accessible only to authorized faculty and staff.

If you have questions regarding this notice or UCalgary's collection, use, or disclosure of your personal information, please contact accessandprivacy@ucalgary.ca.

10.7. Copyright Legislation

All students are required to read the University of Calgary policy on Acceptable Use of Material Protected by Copyright, available at https://www.ucalgary.ca/legal-services/university-policies-procedures/acceptable-use-material-protected-copyright-policy, and requirements of the copyright act, available at https://laws-lois.justice.gc.ca/eng/acts/C-42/index.html, to ensure they are aware of the consequences of unauthorized sharing of course materials (including instructor notes, electronic versions of textbooks etc.). Students who use material protected by copyright in violation of this policy may be disciplined under the Non-Academic Misconduct Policy https://www.ucalgary.ca/legal-services/university-policies-procedures/student-non-academic-misconduct-policy.

10.8. Sexual and Gender-Based Violence Policy

The University recognizes that all members of the University Community should be able to learn, work, teach and live in an environment where they are free from harassment, discrimination, and violence. The University of Calgary's sexual violence policy guides us in how we respond to incidents of sexual violence, including supports available to those who have experienced or witnessed sexual violence, or those who are alleged to have committed sexual violence. It provides clear response procedures and timelines, defines complex concepts, and addresses incidents that occur off campus in certain circumstances. Please see the policy available at https://www.ucalgary.ca/legal-services/university-policies-procedures/sexual-and-gender-based-violence-policy

10.9. Additional Support and Resources

For important information on the following student support and resources, please visit the Registrar's website at: https://www.ucalgary.ca/registrar/registration/course-outlines

- Wellness and Mental Health Resources
- Student Success
- Student Ombuds Office
- Student Union (SU) Information
- Graduate Students' Association (GSA) Information
- Emergency Evacuation/Assembly Points
- Safewalk